

Statistical Interpretation Workflow Library

Expert Workflow Library

Ultimate AI Research Toolkit 2026 - Premium Edition

Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	Research prompts and ideation systems
[Review]	Literature synthesis and gap detection
[Write]	Paper writing and revision workflows
[Plan]	Publication and productivity operations

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Introduction

This Statistical Interpretation collection provides 90 expert research workflows engineered for high-rigor academic and technical delivery. Each workflow includes practical fields to help users move from strategy to publication-ready output.

Instructions

- Select workflows by category and difficulty to match project maturity.
- Replace variables in each workflow with your discipline, data, and resource constraints.
- Use Expected Output as the acceptance criteria for quality control.
- Store outputs in versioned folders to preserve decision history.

Workflow Library

1. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - practical deployment framing

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Effect Size Analysis.

Prompt: Rewrite the methodology around Effect Size Analysis for clinical-risk model evaluation so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for JMLR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=benchmarking specialist. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Strategy

2. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - method comparison clarity

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Write a Nature-style abstract for a study on Uncertainty Quantification in longitudinal causal analysis. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=theory-oriented reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Analysis

3. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - editorial risk reduction

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Robustness Checks.

Prompt: Generate a figure strategy for Robustness Checks with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Pattern Recognition. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=methodology purist. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Writing

4. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - venue-specific optimization

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Hypothesis Testing.

Prompt: Build an ethics checklist and mitigation plan for Hypothesis Testing in small-sample experimental setup. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for Biostatistics review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=interdisciplinary committee member. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Publication

5. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - decision traceability

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Effect Size Analysis.

Prompt: Design a publication strategy for Effect Size Analysis: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for JMLR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=ethics compliance reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Methodology

6. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - citation-strength positioning

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Assume Reviewer #2 challenges Uncertainty Quantification for IEEE JBHI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=reproducibility auditor. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Execution

7. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - defense-ready reasoning

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Robustness Checks.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Robustness Checks. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Pattern Recognition. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=applied domain expert. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Compliance

8. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - high-confidence statistical reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Design a statistical analysis workflow for Hypothesis Testing with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for Biostatistics. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=systems performance reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Reasoning

9. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - evidence-first writing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Effect Size Analysis.

Prompt: Analyze a draft on Effect Size Analysis and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to JMLR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=industry impact reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Analysis

10. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - hypothesis stress testing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Interpret results for Uncertainty Quantification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE JBHI. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=editor focused on clarity. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Writing

11. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - results-to-insight translation

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize rigor and produce publication-grade decisions for Robustness Checks.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Robustness Checks and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Pattern Recognition.

Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=skeptical statistician. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Publication

12. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - major-revision readiness

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize reproducibility and produce publication-grade decisions for Hypothesis Testing.

Prompt: Generate a publication-ready section for Hypothesis Testing tuned for Biostatistics. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=benchmarking specialist. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Methodology

13. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - replication-grade documentation

Objective: Pressure-test experimental design before expensive execution. Prioritize clarity and produce publication-grade decisions for Effect Size Analysis.

Prompt: Audit the experimental design for Effect Size Analysis. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for JMLR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational neuroscience;

dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=theory-oriented reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Execution

14. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - reviewer-resilient argumentation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize statistical power and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Create a LaTeX writing workflow for Uncertainty Quantification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE JBHI submission checks. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=methodology purist. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Compliance

15. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - ethics-by-design reporting

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize transferability and produce publication-grade decisions for Robustness Checks.

Prompt: Simulate a high-pressure Q&A; on Robustness Checks. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=interdisciplinary committee member. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Reasoning

16. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - limitations-first transparency

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize ethical compliance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Act as a senior research editor. Build a contribution architecture for Hypothesis Testing in small-sample experimental setup targeting Biostatistics. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=ethics compliance reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Rebuttal

17. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - cross-domain readability

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize computational efficiency and produce publication-grade decisions for Effect Size Analysis.

Prompt: Rewrite the methodology around Effect Size Analysis for clinical-risk model evaluation so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for JMLR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=reproducibility auditor. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Validation

18. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - impact-focused communication

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize clinical relevance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Write a Nature-style abstract for a study on Uncertainty Quantification in longitudinal causal analysis. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=applied domain expert. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Publication

19. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - publishable novelty framing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize publication readiness and produce publication-grade decisions for Robustness Checks.

Prompt: Generate a figure strategy for Robustness Checks with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Pattern Recognition. Add one caption rewrite for non-expert readers. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=systems performance reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Methodology

20. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - practical deployment framing

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize novelty and produce publication-grade decisions for Hypothesis Testing.

Prompt: Build an ethics checklist and mitigation plan for Hypothesis Testing in small-sample experimental setup. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for Biostatistics review. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=industry impact reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Execution

21. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - method comparison clarity

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize rigor and produce publication-grade decisions for Effect Size Analysis.

Prompt: Design a publication strategy for Effect Size Analysis: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for JMLR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=editor focused on clarity. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Compliance

22. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - editorial risk reduction

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize reproducibility and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Assume Reviewer #2 challenges Uncertainty Quantification for IEEE JBHI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=skeptical statistician. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Reasoning

23. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - venue-specific optimization

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize clarity and produce publication-grade decisions for Robustness Checks.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Robustness Checks. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Pattern Recognition. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=benchmarking specialist. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Rebuttal

24. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - decision traceability

Objective: Select robust statistical strategy and interpretation logic. Prioritize statistical power and produce publication-grade decisions for Hypothesis Testing.

Prompt: Design a statistical analysis workflow for Hypothesis Testing with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for Biostatistics. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=theory-oriented reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Validation

25. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - citation-strength positioning

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize transferability and produce publication-grade decisions for Effect Size Analysis.

Prompt: Analyze a draft on Effect Size Analysis and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to JMLR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=methodology purist. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Optimization

26. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - defense-ready reasoning

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize ethical compliance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Interpret results for Uncertainty Quantification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE JBHI. Add one paragraph that preempts overclaiming. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=interdisciplinary committee member. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Planning

27. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - high-confidence statistical reporting

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize computational efficiency and produce publication-grade decisions for Robustness Checks.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Robustness Checks and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Pattern Recognition. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=ethics compliance reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Execution

28. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - evidence-first writing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize clinical relevance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Generate a publication-ready section for Hypothesis Testing tuned for Biostatistics. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=reproducibility auditor. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Compliance

29. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - hypothesis stress testing

Objective: Pressure-test experimental design before expensive execution. Prioritize publication readiness and produce publication-grade decisions for Effect Size Analysis.

Prompt: Audit the experimental design for Effect Size Analysis. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for JMLR reviewers. Include a stop/go decision protocol. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=applied domain expert. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Reasoning

30. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - results-to-insight translation

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize novelty and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Create a LaTeX writing workflow for Uncertainty Quantification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE JBHI submission checks. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=systems performance reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Rebuttal

31. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - major-revision readiness

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize rigor and produce publication-grade decisions for Robustness Checks.

Prompt: Simulate a high-pressure Q&A; on Robustness Checks. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=industry impact reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Validation

32. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - replication-grade documentation

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize reproducibility and produce publication-grade decisions for Hypothesis Testing.

Prompt: Act as a senior research editor. Build a contribution architecture for Hypothesis Testing in small-sample experimental setup targeting Biostatistics. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=editor focused on clarity. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Optimization

33. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - reviewer-resilient argumentation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize clarity and produce publication-grade decisions for Effect Size Analysis.

Prompt: Rewrite the methodology around Effect Size Analysis for clinical-risk model evaluation so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for JMLR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=skeptical statistician. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Planning

34. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - ethics-by-design reporting

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize statistical power and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Write a Nature-style abstract for a study on Uncertainty Quantification in longitudinal causal analysis. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=benchmarking specialist. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Strategy

35. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - limitations-first transparency

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize transferability and produce publication-grade decisions for Robustness Checks.

Prompt: Generate a figure strategy for Robustness Checks with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Pattern Recognition. Add one caption rewrite for non-expert readers. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=theory-oriented reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Analysis

36. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - cross-domain readability

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize ethical compliance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Build an ethics checklist and mitigation plan for Hypothesis Testing in small-sample experimental setup. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for Biostatistics review. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=methodology purist. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Reasoning

37. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - impact-focused communication

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize computational efficiency and produce publication-grade decisions for Effect Size Analysis.

Prompt: Design a publication strategy for Effect Size Analysis: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for JMLR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=interdisciplinary committee member. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Rebuttal

38. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - publishable novelty framing

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize clinical relevance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Assume Reviewer #2 challenges Uncertainty Quantification for IEEE JBHI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=ethics compliance reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Validation

39. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - practical deployment framing

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize publication readiness and produce publication-grade decisions for Robustness Checks.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Robustness Checks. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Pattern Recognition. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=reproducibility auditor. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Optimization

40. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - method comparison clarity

Objective: Select robust statistical strategy and interpretation logic. Prioritize novelty and produce publication-grade decisions for Hypothesis Testing.

Prompt: Design a statistical analysis workflow for Hypothesis Testing with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for Biostatistics. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=applied domain expert. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from an applied domain expert.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Planning

41. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - editorial risk reduction

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize rigor and produce publication-grade decisions for Effect Size Analysis.

Prompt: Analyze a draft on Effect Size Analysis and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to JMLR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=systems performance reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Strategy

42. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - venue-specific optimization

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize reproducibility and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Interpret results for Uncertainty Quantification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE JBHI. Add one paragraph that preempts overclaiming. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=industry impact reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Analysis

43. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - decision traceability

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize clarity and produce publication-grade decisions for Robustness Checks.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Robustness Checks and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Pattern Recognition.

Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=editor focused on clarity. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Writing

44. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - citation-strength positioning

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize statistical power and produce publication-grade decisions for Hypothesis Testing.

Prompt: Generate a publication-ready section for Hypothesis Testing tuned for Biostatistics. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=skeptical statistician. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Publication

45. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - defense-ready reasoning

Objective: Pressure-test experimental design before expensive execution. Prioritize transferability and produce publication-grade decisions for Effect Size Analysis.

Prompt: Audit the experimental design for Effect Size Analysis. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for JMLR reviewers. Include a stop/go decision protocol. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=benchmarking specialist. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Validation

46. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - high-confidence statistical reporting

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize ethical compliance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Create a LaTeX writing workflow for Uncertainty Quantification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE JBHI submission checks. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=theory-oriented reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Optimization

47. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - evidence-first writing

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize computational efficiency and produce publication-grade decisions for Robustness Checks.

Prompt: Simulate a high-pressure Q&A; on Robustness Checks. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=methodology purist. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Planning

48. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - hypothesis stress testing

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize clinical relevance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Act as a senior research editor. Build a contribution architecture for Hypothesis Testing in small-sample experimental setup targeting Biostatistics. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=interdisciplinary committee member. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Strategy

49. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - results-to-insight translation

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize publication readiness and produce publication-grade decisions for Effect Size Analysis.

Prompt: Rewrite the methodology around Effect Size Analysis for clinical-risk model evaluation so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for JMLR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=ethics compliance reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Analysis

50. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - major-revision readiness

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize novelty and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Write a Nature-style abstract for a study on Uncertainty Quantification in longitudinal causal analysis. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=reproducibility auditor. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Writing

51. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - replication-grade documentation

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize rigor and produce publication-grade decisions for Robustness Checks.

Prompt: Generate a figure strategy for Robustness Checks with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Pattern Recognition. Add one caption rewrite for non-expert readers. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=applied domain expert. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Publication

52. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - reviewer-resilient argumentation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize reproducibility and produce publication-grade decisions for Hypothesis Testing.

Prompt: Build an ethics checklist and mitigation plan for Hypothesis Testing in small-sample experimental setup. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for Biostatistics review. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=systems performance reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Methodology

53. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - ethics-by-design reporting

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize clarity and produce publication-grade decisions for Effect Size Analysis.

Prompt: Design a publication strategy for Effect Size Analysis: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for JMLR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=industry impact reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Execution

54. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - limitations-first transparency

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize statistical power and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Assume Reviewer #2 challenges Uncertainty Quantification for IEEE JBHI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=editor focused on clarity. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Planning

55. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - cross-domain readability

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize transferability and produce publication-grade decisions for Robustness Checks.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Robustness Checks. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Pattern Recognition. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=skeptical statistician. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Strategy

56. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - impact-focused communication

Objective: Select robust statistical strategy and interpretation logic. Prioritize ethical compliance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Design a statistical analysis workflow for Hypothesis Testing with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for Biostatistics. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=benchmarking specialist. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Analysis

57. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - publishable novelty framing

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize computational efficiency and produce publication-grade decisions for Effect Size Analysis.

Prompt: Analyze a draft on Effect Size Analysis and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to JMLR. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=theory-oriented reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Writing

58. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - practical deployment framing

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize clinical relevance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Interpret results for Uncertainty Quantification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE JBHI. Add one paragraph that preempts overclaiming. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=methodology purist;

output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=methodology purist. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Publication

59. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - method comparison clarity

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize publication readiness and produce publication-grade decisions for Robustness Checks.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Robustness Checks and rank them by impact and tractability. Then design three testable gap statements, each with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Pattern Recognition. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=interdisciplinary committee member. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Methodology

60. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - editorial risk reduction

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize novelty and produce publication-grade decisions for Hypothesis Testing.

Prompt: Generate a publication-ready section for Hypothesis Testing tuned for Biostatistics. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=ethics compliance reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Execution

61. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - venue-specific optimization

Objective: Pressure-test experimental design before expensive execution. Prioritize rigor and produce publication-grade decisions for Effect Size Analysis.

Prompt: Audit the experimental design for Effect Size Analysis. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for JMLR reviewers. Include a stop/go decision protocol. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=reproducibility auditor. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Compliance

62. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - decision traceability

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize reproducibility and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Create a LaTeX writing workflow for Uncertainty Quantification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE JBHI submission checks. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=applied domain expert. Example

outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Reasoning

63. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - citation-strength positioning

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize clarity and produce publication-grade decisions for Robustness Checks.

Prompt: Simulate a high-pressure Q&A; on Robustness Checks. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=systems performance reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Analysis

64. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - defense-ready reasoning

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize statistical power and produce publication-grade decisions for Hypothesis Testing.

Prompt: Act as a senior research editor. Build a contribution architecture for Hypothesis Testing in small-sample experimental setup targeting Biostatistics. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=industry impact reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Writing

65. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - high-confidence statistical reporting

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize transferability and produce publication-grade decisions for Effect Size Analysis.

Prompt: Rewrite the methodology around Effect Size Analysis for clinical-risk model evaluation so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for JMLR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=editor focused on clarity. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Publication

66. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - evidence-first writing

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize ethical compliance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Write a Nature-style abstract for a study on Uncertainty Quantification in longitudinal causal analysis. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=skeptical statistician. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Methodology

67. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - hypothesis stress testing

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize computational efficiency and produce publication-grade decisions for Robustness Checks.

Prompt: Generate a figure strategy for Robustness Checks with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Pattern Recognition. Add one caption rewrite for non-expert readers. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=benchmarking specialist. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Execution

68. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - results-to-insight translation

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize clinical relevance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Build an ethics checklist and mitigation plan for Hypothesis Testing in small-sample experimental setup. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for Biostatistics review. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=theory-oriented reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Compliance

69. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - major-revision readiness

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize publication readiness and produce publication-grade decisions for Effect Size Analysis.

Prompt: Design a publication strategy for Effect Size Analysis: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for JMLR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=methodology purist. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Reasoning

70. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - replication-grade documentation

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize novelty and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Assume Reviewer #2 challenges Uncertainty Quantification for IEEE JBHI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=interdisciplinary committee member. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Rebuttal

71. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - reviewer-resilient argumentation

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize rigor and produce publication-grade decisions for Robustness Checks.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Robustness Checks. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Pattern Recognition. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=ethics compliance reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Validation

72. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - ethics-by-design reporting

Objective: Select robust statistical strategy and interpretation logic. Prioritize reproducibility and produce publication-grade decisions for Hypothesis Testing.

Prompt: Design a statistical analysis workflow for Hypothesis Testing with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for Biostatistics. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=reproducibility auditor. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Publication

73. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - limitations-first transparency

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize clarity and produce publication-grade decisions for Effect Size Analysis.

Prompt: Analyze a draft on Effect Size Analysis and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to JMLR. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=applied domain expert. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Methodology

74. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - cross-domain readability

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize statistical power and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Interpret results for Uncertainty Quantification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE JBHI. Add one paragraph that preempts overclaiming. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=systems performance reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Execution

75. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - impact-focused communication

Objective: Detect non-obvious research gaps and convert them into testable opportunities. Prioritize transferability and produce publication-grade decisions for Robustness Checks.

Prompt: You are a literature forensics specialist. From the context below, identify unresolved contradictions in Robustness Checks and rank them by impact and tractability. Then design three testable gap statements, each

with required datasets, threat-to-validity notes, and likely reviewer objections for Elsevier Pattern Recognition. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=industry impact reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Distinguish 'no evidence exists' from 'evidence exists but conflicts' to avoid weak gap claims. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Compliance

76. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - publishable novelty framing

Objective: Produce section drafts compliant with high-impact conference and journal conventions. Prioritize ethical compliance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Generate a publication-ready section for Hypothesis Testing tuned for Biostatistics. Enforce concise technical style, explicit contributions, and strong figure/table callouts. Provide two style variants: IEEE Transactions tone and ACM conference tone with sentence-level rewrites. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=editor focused on clarity. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Avoid generic claims; anchor every paragraph with result-oriented language. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Reasoning

77. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - practical deployment framing - Variant 077

Objective: Pressure-test experimental design before expensive execution. Prioritize computational efficiency and produce publication-grade decisions for Effect Size Analysis.

Prompt: Audit the experimental design for Effect Size Analysis. Produce control/ablation matrix, confound analysis, data leakage checks, minimum sample strategy, and statistical power guardrails suitable for JMLR reviewers. Include a stop/go decision protocol. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=skeptical statistician; output

schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=skeptical statistician. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Design ablations to isolate one causal factor at a time. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Rebuttal

78. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - method comparison clarity - Variant 078

Objective: Convert technical narrative into clean, production-ready LaTeX structure. Prioritize clinical relevance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Create a LaTeX writing workflow for Uncertainty Quantification: section skeleton, theorem/algorithm formatting guidance, table conventions, citation macros, and cross-reference hygiene. Include common formatting errors and fixes for IEEE JBHI submission checks. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=benchmarking specialist. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Draft content in semantic chunks before polishing visual layout. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking specialist.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Validation

79. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - editorial risk reduction - Variant 079

Objective: Prepare rigorous responses for committee or reviewer questioning. Prioritize publication readiness and produce publication-grade decisions for Robustness Checks.

Prompt: Simulate a high-pressure Q&A; on Robustness Checks. Generate 15 hard questions (methods, validity, novelty, limitations, impact) and produce concise, evidence-backed responses with fallback answers for uncertain data points. Include one slide cue per answer. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=theory-oriented reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone

gates.

Tips: Structure answers as claim -> evidence -> limitation -> next action. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Optimization

80. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - venue-specific optimization - Variant 080

Objective: Create a contribution map that defends novelty against closest prior work and defines what is publishable. Prioritize novelty and produce publication-grade decisions for Hypothesis Testing.

Prompt: Act as a senior research editor. Build a contribution architecture for Hypothesis Testing in small-sample experimental setup targeting Biostatistics. Output sections: contribution claim stack, nearest-neighbor prior work comparison, differentiator evidence plan, failure-risk matrix, and a 30/60/90-day execution plan. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=methodology purist; output schema=figure-caption protocol; risk scenario=inter-annotator disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=methodology purist. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Force each claim to link to one measurable piece of evidence and one comparison baseline. Ensure the figure-caption protocol explicitly addresses inter-annotator disagreement and anticipated concerns from a methodology purist.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Planning

81. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - decision traceability - Variant 081

Objective: Rewrite methods for reproducibility and procedural transparency. Prioritize rigor and produce publication-grade decisions for Effect Size Analysis.

Prompt: Rewrite the methodology around Effect Size Analysis for clinical-risk model evaluation so another lab can reproduce it without author contact. Include variable definitions, preprocessing chronology, model hyperparameters, instrumentation assumptions, and reproducibility checklist required for JMLR. Project profile: discipline=computational social science; dataset=open government policy data; preferred method=causal inference pipeline; reviewer persona=interdisciplinary committee member; output schema=statistical decision tree; risk scenario=small sample size constraints; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational social science; dataset=open government policy data; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=interdisciplinary committee member. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Name every tool/version and include randomness-control policy. Ensure the statistical decision tree explicitly addresses small sample size constraints and anticipated concerns from a interdisciplinary committee member.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Execution

82. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - citation-strength positioning - Variant 082

Objective: Craft concise, high-impact abstracts aligned with selective journal standards. Prioritize reproducibility and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Write a Nature-style abstract for a study on Uncertainty Quantification in longitudinal causal analysis. Constrain to problem, approach, principal quantitative finding, boundary condition, and broader implication. Provide a weaker draft and then a stronger revised draft with rationale. Project profile: discipline=materials informatics; dataset=federated hospital dataset; preferred method=Bayesian hierarchical model; reviewer persona=ethics compliance reviewer; output schema=ablation plan sheet; risk scenario=compute budget reduction; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=materials informatics; dataset=federated hospital dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=ethics compliance reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: One central claim per sentence keeps abstract logic clean and persuasive. Ensure the ablation plan sheet explicitly addresses compute budget reduction and anticipated concerns from a ethics compliance reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Compliance

83. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - defense-ready reasoning - Variant 083

Objective: Transform visuals into evidence-rich storytelling assets. Prioritize clarity and produce publication-grade decisions for Robustness Checks.

Prompt: Generate a figure strategy for Robustness Checks with publication-grade caption drafts. For each figure, define purpose, metric, axis design, uncertainty communication, and interpretation sentence expected by Elsevier Pattern Recognition. Add one caption rewrite for non-expert readers. Project profile: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; preferred method=graph neural workflow; reviewer persona=reproducibility auditor; output schema=rebuttal response grid; risk scenario=time-limited resubmission window; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=computational neuroscience; dataset=crowdsourced annotation dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer

persona=reproducibility auditor. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Good captions stand alone and state what changed, by how much, and why it matters. Ensure the rebuttal response grid explicitly addresses time-limited resubmission window and anticipated concerns from a reproducibility auditor.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Reasoning

84. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - high-confidence statistical reporting - Variant 084

Objective: Embed ethics and compliance checks into the technical workflow. Prioritize statistical power and produce publication-grade decisions for Hypothesis Testing.

Prompt: Build an ethics checklist and mitigation plan for Hypothesis Testing in small-sample experimental setup. Include fairness risks, privacy boundaries, human-subject concerns, reporting transparency, and audit evidence expected for Biostatistics review. Project profile: discipline=human-computer interaction; dataset=industrial telemetry stream; preferred method=mixed-methods protocol; reviewer persona=applied domain expert; output schema=section rewrite pack; risk scenario=ethics board clarification request; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=human-computer interaction; dataset=industrial telemetry stream; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=applied domain expert. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Translate ethical risks into actionable controls with owner and verification evidence. Ensure the section rewrite pack explicitly addresses ethics board clarification request and anticipated concerns from a applied domain expert.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Rebuttal

85. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - evidence-first writing - Variant 085

Objective: Choose venue strategy, narrative framing, and revision sequencing. Prioritize transferability and produce publication-grade decisions for Effect Size Analysis.

Prompt: Design a publication strategy for Effect Size Analysis: venue ladder (A-plan/B-plan), novelty framing options, deadline-aligned writing sprints, and reviewer-risk scenarios. Include a branch for conference-first then journal-extension path for JMLR. Project profile: discipline=robotics; dataset=city-scale mobility records; preferred method=time-series anomaly framework; reviewer persona=systems performance reviewer; output schema=action timeline; risk scenario=conflicting baseline outcomes; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=robotics; dataset=city-scale mobility records; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=systems performance reviewer. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Plan fallback venue narratives early to reduce resubmission cycle time. Ensure the action timeline explicitly addresses conflicting baseline outcomes and anticipated concerns from a systems performance reviewer.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Validation

86. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - hypothesis stress testing - Variant 086

Objective: Turn skeptical reviewer criticism into a persuasive, evidence-linked response workflow. Prioritize ethical compliance and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Assume Reviewer #2 challenges Uncertainty Quantification for IEEE JBHI. Draft a response package with: acknowledgement sentence, technical clarification, direct manuscript change, optional added experiment, and fallback wording if resources are limited. Provide line-level mapping format for revision tracking. Project profile: discipline=digital health; dataset=public genomics panel; preferred method=hybrid rule-neural model; reviewer persona=industry impact reviewer; output schema=risk register; risk scenario=missing metadata in source data; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=digital health; dataset=public genomics panel; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=industry impact reviewer. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Use respectful tone and separate explanation, action, and manuscript delta into distinct bullets. Ensure the risk register explicitly addresses missing metadata in source data and anticipated concerns from a industry impact reviewer.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Optimization

87. Audit Statistical Validity: effect-size narrative workflow for imbalanced-class benchmarking (Elsevier Pattern Recognition) - results-to-insight translation - Variant 087

Objective: Plan a high-confidence major revision package for journal resubmission. Prioritize computational efficiency and produce publication-grade decisions for Robustness Checks.

Prompt: Create a major revision plan for an Elsevier-style decision letter focused on Robustness Checks. Include comment clustering, evidence plan, timeline by dependency, added-analysis options, and response-letter structure that anticipates editor priorities for Elsevier Pattern Recognition. Project profile: discipline=machine learning; dataset=benchmark tabular risk dataset; preferred method=multi-objective optimizer; reviewer persona=editor focused on clarity; output schema=claim-evidence table; risk scenario=incomplete ablation coverage; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=machine learning; dataset=benchmark tabular risk dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Elsevier Pattern Recognition; reviewer persona=editor focused on clarity. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Group reviewer comments by technical dependency to prevent contradictory edits. Ensure the claim-evidence table explicitly addresses incomplete ablation coverage and anticipated concerns from a editor focused on clarity.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Robustness Checks.

Difficulty: Advanced

Category: Planning

88. Plan Statistical Analysis Like a Reviewer: test-selection decision tree for small-sample experimental setup (Biostatistics) - major-revision readiness - Variant 088

Objective: Select robust statistical strategy and interpretation logic. Prioritize clinical relevance and produce publication-grade decisions for Hypothesis Testing.

Prompt: Design a statistical analysis workflow for Hypothesis Testing with assumptions, test-selection logic, effect-size reporting, multiplicity correction, and uncertainty interpretation. Output a decision tree that maps data conditions to recommended tests for Biostatistics. Project profile: discipline=climate informatics; dataset=multi-modal image-text corpus; preferred method=semi-supervised training setup; reviewer persona=skeptical statistician; output schema=decision matrix; risk scenario=dataset shift at deployment; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=climate informatics; dataset=multi-modal image-text corpus; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=Biostatistics; reviewer persona=skeptical statistician. Example outcome: result interpretation grid with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Pair p-values with effect sizes and confidence intervals in every result claim. Ensure the decision matrix explicitly addresses dataset shift at deployment and anticipated concerns from a skeptical statistician.

Expected Output: A publication-ready result interpretation grid that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Hypothesis Testing.

Difficulty: Expert

Category: Strategy

89. Build an Uncertainty-First Results Section: uncertainty communication framework for clinical-risk model evaluation (JMLR) - replication-grade documentation - Variant 089

Objective: Detect weak citation support and strengthen scholarly positioning. Prioritize publication readiness and produce publication-grade decisions for Effect Size Analysis.

Prompt: Analyze a draft on Effect Size Analysis and identify citation gaps by claim type: foundational, comparative, methodological, and limitation-related. Return a prioritized fill plan with query phrases, likely source types, and insertion points tailored to JMLR. Project profile: discipline=cybersecurity; dataset=longitudinal sensor dataset; preferred method=counterfactual evaluation approach; reviewer persona=benchmarking specialist; output schema=compliance checklist; risk scenario=unexpected reviewer disagreement; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=cybersecurity; dataset=longitudinal sensor dataset; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=JMLR; reviewer persona=benchmarking specialist. Example outcome: evidence summary with risk controls, reviewer-facing rationale, and milestone gates.

Tips: Flag unsupported superlatives first; they trigger avoidable reviewer criticism. Ensure the compliance checklist explicitly addresses unexpected reviewer disagreement and anticipated concerns from a benchmarking

specialist.

Expected Output: A publication-ready evidence summary that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Effect Size Analysis.

Difficulty: Beginner

Category: Analysis

90. Strengthen Significance Claims: robustness audit protocol for longitudinal causal analysis (IEEE JBHI) - reviewer-resilient argumentation - Variant 090

Objective: Convert raw metrics into defensible scientific interpretation. Prioritize novelty and produce publication-grade decisions for Uncertainty Quantification.

Prompt: Interpret results for Uncertainty Quantification using a claim-evidence map: primary findings, uncertainty envelope, competing explanations, boundary conditions, and implications for future work in IEEE JBHI. Add one paragraph that preempts overclaiming. Project profile: discipline=biomedical engineering; dataset=multi-site clinical cohort; preferred method=transformer baseline; reviewer persona=theory-oriented reviewer; output schema=citation repair map; risk scenario=high class imbalance; timeline={timeline}; resources={resources}. Return outputs as numbered sections and end with an execution checklist.

Example: Example input: discipline=biomedical engineering; dataset=multi-site clinical cohort; timeline=14 weeks; resources=3 researchers + 2 GPUs; target venue=IEEE JBHI; reviewer persona=theory-oriented reviewer. Example outcome: statistical memo with risk controls, reviewer-facing rationale, and milestone gates.

Tips: State what the results do not prove as clearly as what they do prove. Ensure the citation repair map explicitly addresses high class imbalance and anticipated concerns from a theory-oriented reviewer.

Expected Output: A publication-ready statistical memo that includes decisions, rationale, measurable criteria, and reviewer-resilient framing for Uncertainty Quantification.

Difficulty: Intermediate

Category: Reasoning

Summary

Applying these 90 workflows systematically improves decision quality, writing clarity, and publication readiness across research cycles.